SGT

SGT HYDROEDGE | GREENLABS

# Hydrogen Fuel Assist for Industrial Boilers

*Cut fuel costs. Slash emissions. Any fuel type. Any boiler size.*

## THE PROBLEM

Industrial boilers consume enormous quantities of diesel, coal, biomass, and gas — yet most operate below peak thermal efficiency. Incomplete combustion wastes fuel, generates excess CO2, and produces harmful particulate matter, CO, NOx, and SOx. With tightening emission norms and ESG mandates, the cost of inefficiency is no longer just fuel waste — it is regulatory risk and lost competitiveness.

## THE SGT HYDROEDGE SOLUTION

SGT HydroEdge is the first Indian company to manufacture high-capacity HHO (oxyhydrogen) generators purpose-built for industrial boilers. Our system generates hydrogen-oxygen gas on-demand through water electrolysis and injects it into the boiler's combustion chamber. Hydrogen's wide flammability range (4–75% in air) and extremely low ignition energy (0.02 mJ) enable faster, more complete combustion of the primary fuel — regardless of whether it is diesel, coal, biomass, or gas.

## SGT'S PATENT-APPLIED TECHNOLOGY ADVANTAGE

SGT HydroEdge has developed proprietary, patent-applied HHO generator technology that goes far beyond conventional electrolysers. Our systems are engineered specifically for industrial-scale combustion applications:

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|---|---|
| **High HHO Yield** | Patent-applied cell design delivers significantly higher hydrogen-oxygen gas output per unit of electrical input compared to conventional electrolysers |
| **Low Power Consumption** | Optimised electrode geometry and electrolyte management reduce electricity consumption, ensuring the net energy benefit is maximised |
| **Long Operational Life** | Engineered for continuous industrial duty. Durable & cost effective materials and design choices ensure years of reliable operation without performance degradation |
| **Safety by Design** | On-demand generation means zero hydrogen storage. Built-in pressure relief, flashback arrestors, and fail-safe controls. No explosive gas inventory on site. |
| **Easy & Low-Cost Maintenance** | Modular design enables quick servicing. No specialised tools or technicians required. Consumables are water and standard electrolyte — both inexpensive and widely available. |

## DIGITALLY CONNECTED — IOT & DIGITAL TWIN READY

**Every SGT HydroEdge system is fully digitally connected.**

Built-in IoT sensors continuously monitor system performance, HHO output, power consumption, and boiler operating parameters. This data feeds directly into GreenVision — SGT's proprietary AI-powered Digital Twin platform.

- **Real-time monitoring** of HHO system health, gas output, and boiler efficiency gains
- **Emission tracking** — measurement and estimation of CO2, PM, CO, NOx, and SOx reductions
- **ESG-grade reporting** — automated reports documenting your emission reductions
- **Predictive maintenance** — AI detects performance degradation before it becomes a problem.

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## WHAT GLOBAL RESEARCH PROVES

Researchers across the world have extensively studied HHO-enhanced combustion in boilers across every major fuel type. The results consistently demonstrate measurable improvements in efficiency and significant reductions in emissions. Here is what the published research shows:

### COAL-FIRED BOILERS

- Adding HHO gas to coal combustion improves boiler efficiency by enabling more complete burning of carbon content in the coal
- Each kilogram of HHO gas introduced saves approximately 5.4 kg of sub-bituminous coal and 7.9 kg of Indian lignite — a significant fuel reduction
- Testing conducted on a 40 THP fluidized bed combustion boiler at an industrial facility in Surat, India confirmed efficiency gains and emission reductions
- In coal-fired thermal power stations, introducing oxyhydrogen into the furnace air intake reduced NOx emissions from 300 mg/m³ to below 100 mg/m³ — a reduction exceeding 66%
- HHO-assisted combustion at ultra-low boiler loads (down to 20%) maintained stable combustion while reducing CO and SO2 concentrations in flue gas

### BIOMASS-FIRED BOILERS

- When HHO gas was introduced into a biomass hot air generator, CO emissions decreased by 93%, NO emissions decreased by 22.5%, and visible smoke decreased by 80%
- Total fuel cost during operation was reduced despite the electricity consumption of the HHO generator — confirming net economic benefit
- Biomass combustion efficiency improvements in the range of 10% to 20% have been documented depending on the existing hardware and operating conditions
- The dramatically reduced smoke levels from the chimney provide visible, immediate evidence of improved combustion to plant operators and regulators

### OIL & DIESEL-FIRED BOILERS

- Combining HHO with heavy oil (HO) in boilers increased steam production by 4% to 27% depending on the quantity of HHO introduced — directly increasing boiler output without additional fuel
- Emission levels of multiple pollutant categories (CO, PM, HC, SOx) were reduced across all tested oil-HHO combinations
- Oil-fired boiler combustion became visibly more complete with HHO supplementation, evidenced by reduced soot, cleaner exhaust, and lower stack temperatures
- Diesel-fired boilers showed improved combustion efficiency due to hydrogen's faster flame propagation speed and catalytic effect on hydrocarbon combustion

### NATURAL GAS-FIRED BOILERS

- Studies on industrial steam boilers up to 108 tonnes per hour and 2.8 MW capacity confirmed that hydrogen-enriched fuel improves thermal efficiency
- Efficiency improvements of 1.2% to 1.6 percentage points documented across multiple studies when hydrogen is blended with natural gas. Increased HHO has potential to improve further efficiency.
- Hydrogen enrichment increases heat transfer through enhanced convection and flue gas radiation, with additional latent heat recovery in condensing boilers
- CO2 emissions decreased by up to 16% when hydrogen content was increased to 40% by volume in natural gas-fired boilers

*These findings are based on research conducted by scientists and engineers at leading institutions worldwide including Tsinghua University (China), CanmetENERGY / Natural Resources Canada, L&T Industries (India), Sapienza University of Rome (Italy), and multiple research groups publishing in peer-reviewed journals indexed in ScienceDirect, MDPI, Elsevier, and Springer. Full research references are available on request.*

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# SGT 3000 LPH – HHO Generator for Boilers


**High-Efficiency | Modular | Made in India**

- Output: Up to 3000 LPH HHO, on-demand & safe
- Technology: Liquid Cell – high efficiency, low maintenance
- Scalable: Modular design for higher HHO requirement
- Power: 10 kVA IGBT-based high-efficiency supply
- Operation: 24×7 continuous, up to 7 days non-stop
- Safety: Multiple electrical & process interlocks
- Retrofit: No boiler or fuel system modification

Designed, developed & manufactured by SGT HydroEdge in Pune, India

**Application: Industrial boilers, Kilns & thermal systems**

## AT A GLANCE

| 2–12% | Up to 93% | Up to 80% | Up to 66% | Up to 27% |
|---|---|---|---|---|
| Fuel Efficiency Improvement | CO Reduction | PM & Smoke Reduction | NOx Reduction | Steam Output Increase |

| | |
|---|---|
| **Fuel types supported** | Diesel, coal (all grades), biomass, natural gas, HFO, LDO |
| **Boiler sizes** | Small industrial to large utility-scale |
| **Installation** | Retrofit. No boiler modification. Minimal downtime. |
| **Hydrogen storage** | None — on-demand generation. Zero storage risk. |
| **Connectivity** | IoT enabled. Integrates with GreenVision Digital Twin. |
| **Technology** | Patent-applied. High yield. Low power. Long life. |

## Ready to Cut Your Boiler Fuel Costs & Emissions?

Ask us for a no-obligation assessment of your boiler operations.

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*SGT HydroEdge — Sustainability Infrastructure for Industry*